

- Developed a secure and scalable E-commerce backend system using Spring Boot, implementing JWT-based authentication and role-based authorization.
- Designed and implemented RESTful APIs for product management, cart operations, order processing, and payment integration.
- Implemented user authentication using Spring Security and JWT, ensuring secure access control for ADMIN and USER roles.
- Developed product management module with CRUD operations and validation.
- Built cart functionality allowing users to add, update, and remove items, ensuring user-specific data isolation.
- Designed order processing system to convert cart items into orders with total price calculation and status tracking.
- Integrated Stripe payment gateway to handle secure payment processing and updated order status upon successful payment.
- Used Spring Data JPA for database interaction with proper entity relationships (One-to-Many and Many-to-Many).
- Implemented global exception handling and validation for robust API responses.
- Tested all APIs using Postman and ensured end-to-end functionality from user registration to payment completion.

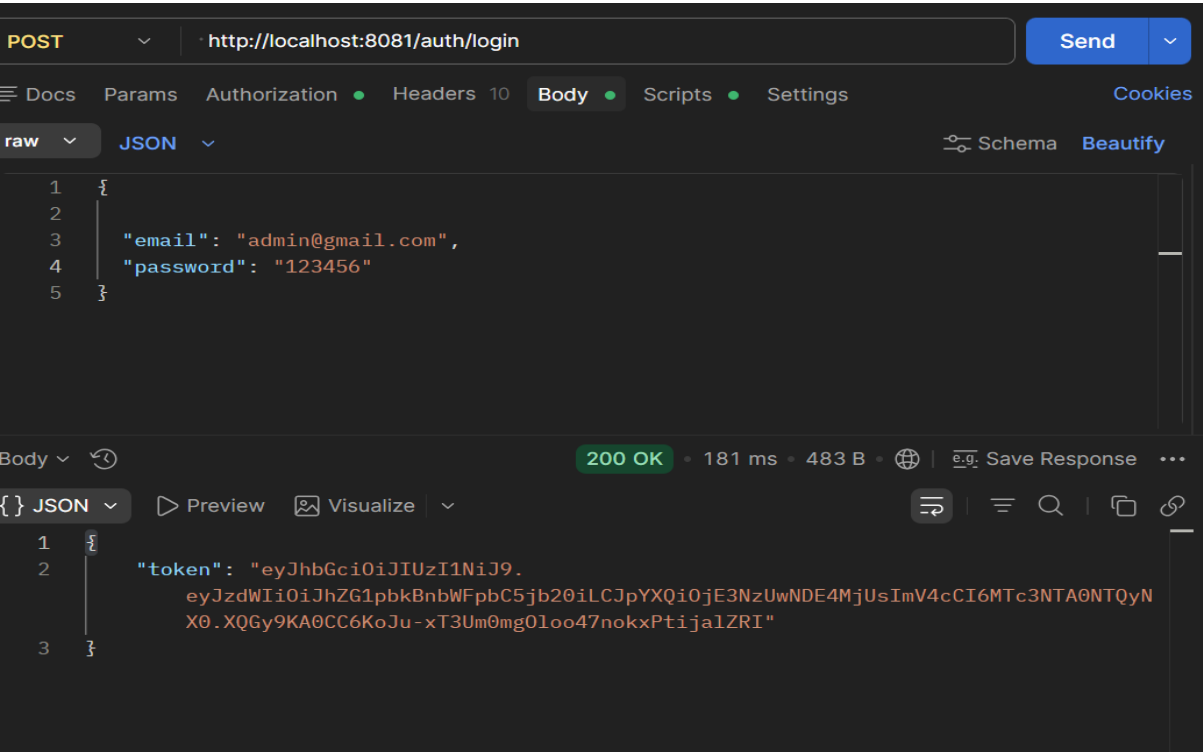
Register (Admin)

POST /auth/register

Login

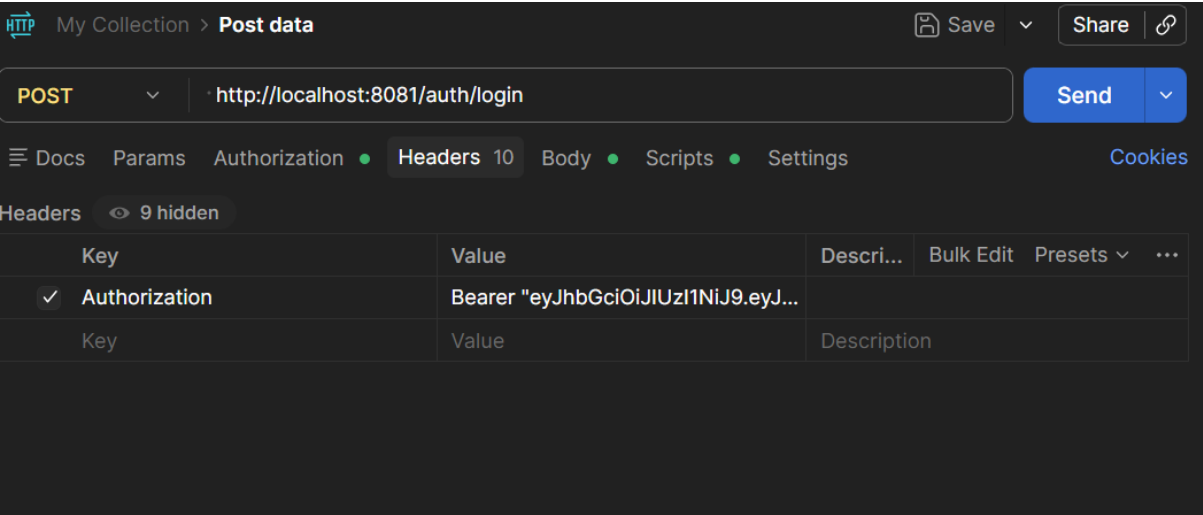
POST /auth/login

Token Gives



Authorization

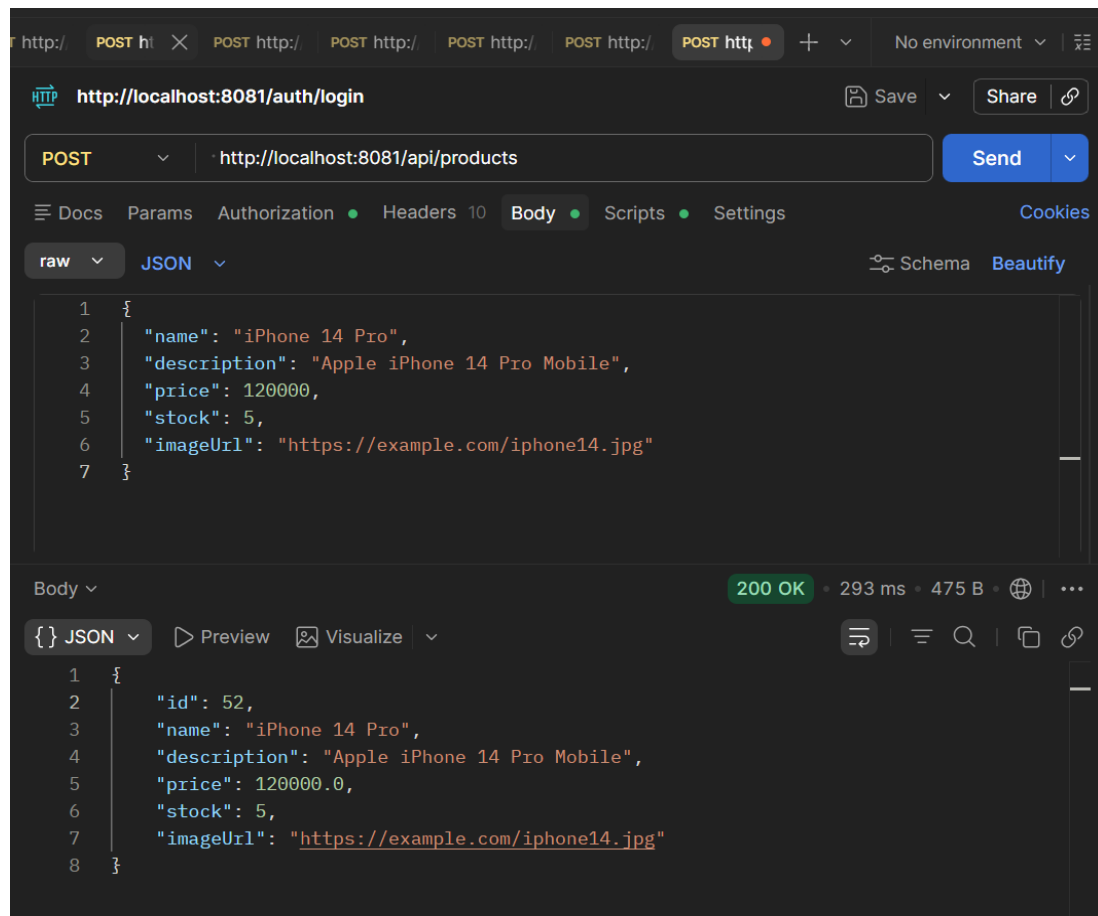
Bearer Token set



Product APIs

Admin login

Product add



Register (Admin)

POST /auth/register

Login

POST /auth/login

Token Gives

View product

HTTP

http://localhost:8081/auth/login

Save

Share

GET

http://localhost:8081/api/products

Send

Docs

Params

Authorization

Headers 10

Body

Scripts

Settings

Cookies

Headers

9 hidden

Key	Value	Descri...	Bulk Edit	Presets	...
✓ Authorization	Bearer eyJhbGciOiJIUzI1NiJ9.eyJz...				
Key	Value	Description			

Body

200 OK • 273 ms • 679 B •

JSON

Preview

Visualize

```
1  [
2    {
3      "id": 1,
4      "name": "Laptop",
5      "description": "Gaming Laptop",
6      "price": 70000.0,
7      "stock": 5,
8      "imageUrl": null
9    },
10  ]
```

```
17  },
18  {
19    "id": 52,
20    "name": "iPhone 14 Pro",
21    "description": "Apple iPhone 14 Pro Mobile",
22    "price": 120000.0,
23    "stock": 5,
24    "imageUrl": "https://example.com/iphone14.jpg"
25  }
26  ]
```

Cart APIs

- Add to cart
- View cart

The screenshot shows a REST client interface with the following details:

- URL:** `http://localhost:8081/auth/login`
- Method:** `POST`
- Path:** `http://localhost:8081/api/cart/add?productId=1&quantity=2`
- Headers:** A table with one entry:

Key	Value	Description
Authorization	Bearer eyJhbGciOiJIUzI1NiJ9.eyJz...	
- Body:** A JSON object:

```
{  "user": {    "id": 52,    "name": "User1",    "email": "user1@gmail.com",    "password": "$2a$10$nILsS/mwr3Uek0180Qc3I09b5WCDGBY0gBH0GzJ0kX1qe8pfUfSmG",    "role": "ROLE_USER"  },  "product": {    "id": 1,    "name": "Laptop",    "description": "Gaming Laptop",    "price": 70000.0,    "stock": 5,    "imageUrl": null  },  "quantity": 2}
```
- Status:** `200 OK` (100 ms, 606 B)
- Response Format:** JSON

Order APIs

- Create order
- View orders

Payment APIs

- Create payment (Stripe)
- Payment success → status updated

POST http://localhost:8081/api/orders/create

Send

Docs Params Authorization Headers 10 Body Scripts Settings Cookies

Key	Value	Descr...	Bulk Edit	Presets	...
✓ Authorization	Bearer eyJhbGciOiJIUzI1NiJ9.eyJz...				
Key	Value	Description			

Body

200 OK • 157 ms • 674 B • 🌐 | ...

{ } JSON Preview Visualize

```
7   "password": "$2a$10$ILsS/mwr3Uek0180Qc3I09b5WCDGBY0gBH0GzJ0kX1qe8pfUfSmG",
8   "role": "ROLE_USER"
9 },
10  "products": [
11    {
12      "id": 1,
13      "name": "Laptop",
14      "description": "Gaming Laptop",
15      "price": 70000.0,
16      "stock": 5,
17      "imageUrl": null
18    }
19  ],
20  "totalAmount": 140000.0,
21  "status": "PENDING",
22  "shippingAddress": "Default Address"
23 }
```

POST http://localhost:8081/auth/login

Save Share

POST http://localhost:8081/payment/create?amount=140000

Send

Docs Params Authorization Headers 10 Body Scripts Settings Cookies

Key	Value	Descr...	Bulk Edit	Presets	...
✓ Authorization	Bearer eyJhbGciOiJIUzI1NiJ9.eyJz...				
Key	Value	Description			

Body

200 OK • 1.13 s • 466 B • 🌐 | ...

{ } JSON Preview Visualize

```
1  {
2    "amount": 14000000,
3    "clientSecret": "pi_3THQiX8UxTgo3GrV0s2UrVf5_secret_t06TBxDIIg1FJ7ycDn40TRowR",
4    "id": "pi_3THQiX8UxTgo3GrV0s2UrVf5"
5  }
```